

Phase 4: Project Planning Phase

The Smart Lender project was planned and executed over a 4-week schedule during the internship period. Each week was assigned a clear focus area, moving progressively from data preparation to model building, backend/API development, and finally integration and testing. The week-wise breakdown is summarized in the table below.

Week	Focus Area	Planned Tasks
Week 1	Environment Setup & Data Collection	Set up Python environment, install dependencies (Flask, Scikit-Learn, XGBoost, Pandas). Collect and load the loan approval dataset. Perform initial data cleaning (whitespace, column names).
Week 2	Exploratory Data Analysis & Model Building	Perform EDA (summary statistics, correlation heatmap, class distribution) in Jupyter Notebook. Encode categorical features and scale numerical features. Train and compare Logistic Regression, Random Forest, and XGBoost models.
Week 3	Backend & API Development	Select the best-performing model and save it with the scaler using Joblib. Build the Flask backend with a /predict REST API endpoint. Develop the responsive HTML/CSS/JavaScript frontend and connect it to the API.
Week 4	Integration, Validation & Final Testing	Integrate frontend and backend, test the complete prediction flow end-to-end. Validate model outputs against known test cases. Fix bugs, refine UI, and prepare project documentation and demonstration.